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Exploratory Landscape Analysis for Meta-Learning-Based Algorithm Selection in Continuous Optimization

Vasileios Charilogis[1], Ioannis G. Tsoulos[2,*] and Anna Maria Gianni[3]

[1] Department of Informatics and Telecommunications, University of Ioannina, 47150 Kostaki Artas, Greece, v.charilog@uoi.gr
[2] Department of Informatics and Telecommunications, University of Ioannina, 47150 Kostaki Artas, Greece, itsoulos@uoi.gr
[3] Department of Informatics and Telecommunications, University of Ioannina, 47150 Kostaki Artas, Greece, am.gianni@uoi.gr
* Correspondence: itsoulos@uoi.gr

Abstract: Continuous optimization now draws on a large and growing number of competing metaheuristic families, including differential-evolution lineages, covariance-matrix adaptation strategies, swarm-intelligence methods, and simulated annealing, yet no single method dominates across problem classes, a consequence of the No Free Lunch theorem. This motivates algorithm selection: choosing, for an unseen problem, the method most likely to perform best. Building on the Optim-Solution framework, we assemble a headless, reproducible batch-execution pipeline coupled with Exploratory Landscape Analysis (ELA), enabling large-scale, statistically grounded benchmarking together with feature-based meta-learning. We evaluate a set of optimizers of comparable overall strength, drawn from distinct algorithmic families, across a broad benchmark suite spanning classical scalable functions and established competition test suites, under a consistent multi-run evaluation protocol. A Random Forest classifier trained on ELA features, validated under a Leave-One-Problem-Out cross-validation scheme, is shown to predict the best-performing optimizer for a previously unseen problem substantially more accurately than the naive Single Best Solver (SBS) baseline, closing a meaningful share of the gap toward the oracle Virtual Best Solver (VBS). Family-level analysis further shows that structured competition benchmarks and classical scalable functions tend to be won by different solvers, confirming that the underlying selection problem is genuine rather than trivial. These results indicate that landscape-aware selection can meaningfully outperform naive strategies when the candidate methods are drawn from heterogeneous algorithmic families of comparable strength.

Keywords: Exploratory Landscape Analysis; Continuous optimization; Random Forest Classifier; Decision Tree; Metaheuristics; Algorithm Selection; Algorithm benchmarking;

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1. Introduction

Continuous global optimization is a core computational task across nearly every quantitative discipline, from engineering design to the tuning of machine-learning models. Over the past three decades the algorithmic community's response to this need has been extraordinarily prolific: alongside classical evolutionary algorithms [1], the literature now offers entire families of population-based metaheuristics with distinct search mechanisms. Differential Evolution (DE) [2] gave rise to a long lineage of adaptive variants built around success-history parameter control and archive-based mutation, including JADE [3], SHADE [4], L-SHADE [5], and jSO [6]. Evolution Strategies gave rise to the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [7], which adapts a full covariance model of the search distribution rather than perturbing individual variables independently. Swarm intelligence contributed Particle Swarm Optimization [8], Ant Colony Optimization for continuous domains [9], and, more recently, nature-inspired variants such as the Grey Wolf

Optimizer [10]. Physics-inspired approaches such as Simulated Annealing [11] remain in wide use as a comparatively low-cost baseline. Each of these families embodies a different bias about how the search space should be explored and exploited, and new variants within each family continue to appear every year, making the practical choice of "which method to use" increasingly difficult rather than easier.

This proliferation is not incidental. The No Free Lunch theorem for optimization [12] formally establishes that, averaged over the space of all possible objective functions, no single optimizer can outperform any other: elevated performance on one class of problems is necessarily offset by inferior performance elsewhere. In practice this means that the "best" algorithm for a given task is contingent on the structure of the problem itself: its modality, separability, ruggedness, and the presence of plateaus or deceptive basins of attraction. The theorem's averaging is taken over the entire, largely unstructured space of conceivable objective functions, so it does not preclude one method from being reliably better on the much narrower, structured subset of problems that arise in practice. It does, however, remove any theoretical guarantee that a single method will do so, which is precisely why an empirical, data-driven selection procedure of the kind developed here has scientific value. The natural response is therefore not to search for a universally superior algorithm, but to select, for a given unseen problem, the method most likely to perform well on it: the algorithm selection problem, originally formalized by Rice [13] and later surveyed extensively from a meta-learning perspective [14], [37]. Exploratory Landscape Analysis (ELA) [15] operationalizes this idea for continuous black-box optimization: a modest sample of the search space is used to compute a compact set of numerical descriptors, related to convexity, level-set structure, local-versus-global information, and similar properties, that summarize how difficult or structured a landscape is, without requiring the objective function to be evaluated exhaustively. Coupled with supervised machine learning, ELA features have been shown capable of predicting which solver from a portfolio is likely to perform best on a new problem instance, closing part of the gap between naive and oracle solver selection [16], [37], [17], [18], and this line of work has been placed within the broader landscape of algorithm-selection methods for black-box continuous optimization in dedicated surveys [19], [34], [35].

Realizing this landscape-aware selection pipeline in practice is largely an infrastructural problem: it requires (i) a means of running many (method, problem, dimension) combinations reproducibly, (ii) a statistically sound procedure for deciding which method actually "won" each combination, and (iii) an integrated mechanism for sampling problem landscapes and extracting ELA features at scale. In this work we build on the OptimSolution framework [20] and assemble around it a headless, batch-execution pipeline coupled with an ELA extraction stage, producing large, reproducible meta-optimization datasets through four stages: batch execution and statistical ranking, including Friedman omnibus testing [25] and Holm-corrected [26] pairwise Wilcoxon signed-rank tests [27], following established practice for comparing evolutionary and swarm-intelligence algorithms [28], [38], [29]. The pipeline then performs landscape sampling, ELA feature computation [15], and, finally, trains a landscape-aware selection model with a Random Forest classifier [30].

The purpose of this paper is to demonstrate, using this pipeline, that a machine-learning model trained on landscape features can predict which optimizer will perform best on a previously unseen problem more accurately than the simplest sensible baseline, always recommending the single method with the best average performance overall, or Single Best Solver (SBS), and to do so with a degree of rigor that makes the comparison meaningful. For this comparison to be genuine rather than trivial, the candidate methods must be of comparable overall strength: pairing one markedly strong solver against several markedly weaker baselines collapses the meta-learning task into a near single-label classification problem, in which no model, however well designed, can outperform the naive SBS baseline by a meaningful margin. We therefore evaluate eight optimizers of comparable overall strength, drawn from four distinct algorithmic families (DE-lineage, CMA-ES, swarm intelligence, and simulated annealing), over 97 (problem, dimension) instances spanning

classical scalable functions and the CEC2017 [31] and CEC2022 [32] competition benchmark suites. The novelty of this work is therefore threefold: (i) an explicit methodological argument, together with its empirical confirmation, that the algorithm-selection problem is only genuine, and only worth learning, when the candidate methods are of comparable strength and drawn from heterogeneous algorithmic families. (ii) a large, reproducible meta-optimization dataset assembled around the OptimSolution framework [20], spanning 97 (problem, dimension) instances and eight optimizers. (iii) a statistically rigorous evaluation of the resulting landscape-aware model, benchmarked under Leave-One-Problem-Out cross-validation against both the naive SBS baseline and the oracle Virtual Best Solver (VBS), quantifying not just whether landscape-awareness helps but how much of the achievable performance gap it actually closes.

The remainder of this paper is organized as follows. Section 2 describes the methodological pipeline built around the OptimSolution framework, covering batch execution and Exploratory Landscape Analysis, at the level of what it enables scientifically. Section 3 details the experimental design adopted for the large-scale study: the eight optimizers and their algorithmic families, the benchmark problems drawn from classical, CEC2017, and CEC2022 suites, and the evaluation protocol. Section 4 presents the results of the experiment: per-method statistical rankings, family-level specialization patterns across benchmark suites, and the Leave-One-Problem-Out accuracy of the landscape-aware selection model relative to the SBS and VBS baselines. Section 5 discusses the practical implications and limitations of these findings, including the effect of label tie-breaking on the reported results. Section 6 concludes the paper.

2. Framework Extensions: Headless Batch Execution and Landscape Analysis

Realizing the landscape-aware selection pipeline motivated in Section 1 required extending the OptimSolution framework [20] along three complementary axes: a way to execute large numbers of (method, problem, dimension) combinations unattended and reproducibly, a principled statistical procedure for turning raw per-combination performance into a labelled meta-dataset, and a mechanism for sampling each problem's landscape and exporting it for downstream feature extraction. This section describes what each addition does at the methodological level.

Batch execution is driven by a compact specification that defines, for a given experiment, the optimizers, benchmark problems, and dimensions to combine. Two complementary specification styles are supported: a full cartesian product over methods, problems, and dimensions, and an explicit, sparse list of individual (method, problem, dimension) triples, the latter allowing an existing experiment to be extended incrementally without re-running combinations already completed. Each combination is executed as an independent, seeded run producing a standardized per-combination summary. These summaries are subsequently merged, by column name rather than by raw file order, into a single master table. This master table then passes through a statistical ranking stage that, for every (problem, dimension) group, determines each method's rank, identifies the group's best-performing method, and tests whether the differences between methods are large enough to be meaningful: a Friedman omnibus test across all methods, followed by pairwise Wilcoxon signed-rank tests with Holm correction for multiple comparisons. The Friedman test is a non-parametric analogue of repeated-measures analysis of variance: it ranks the methods separately within every (problem, dimension) group and asks whether the resulting average ranks differ more than would be expected by chance alone. When it indicates that they do, the Wilcoxon signed-rank test is used to compare two methods at a time, by ranking the magnitude of their paired performance differences across groups, and the Holm correction lowers the significance threshold required of each individual pairwise test to control the overall risk of a false positive that grows as more pairs are tested simultaneously. It is this ranking stage that turns raw performance numbers into the labelled meta-dataset used to train the selection model evaluated in Section 5 (Tables 2 and 3).

In parallel, a landscape-sampling stage draws a fixed budget of points from each problem's true, declared search bounds, using the same initialization procedure, uniform, Latin-hypercube, or Halton, already used to seed every optimizer in the framework, so that the sample is drawn under exactly the assumptions the optimizers themselves operate under. The problem's real objective function, not a surrogate, is evaluated at each sampled point. Sample size follows the convention established in the Exploratory Landscape Analysis literature [15] of tens to a few hundred points per dimension: the pipeline draws exactly 100 samples per problem dimension for every instance, using each problem's declared dimension as listed in Table 1. For the classical Powell function this declared dimension is not the same one the optimizers themselves run at, as Section 3 explains and Section 5 discusses further. The resulting points and their function values, together with each problem's declared modality and separability, are exported for feature extraction. Computing the actual ELA descriptors from this sample, distributional statistics, meta-model fit quality, information-content and nearest-better-clustering measures, and level-set classification errors, corresponding to the feature families reported in Table 4, is deliberately delegated to the established pflacco library [33] rather than re-implemented, so that the study relies on independently validated statistics [36] rather than a bespoke reimplementation.

The labelled performance table and the extracted landscape features are finally joined on (problem, dimension) and used to train a Random Forest classifier [30]. A Random Forest is an ensemble learner that trains a large number of individual decision trees, each on an independently resampled subset of the training data and a randomly restricted subset of the available features, and combines their predictions by majority vote. A single decision tree classifies an instance by applying a sequence of threshold tests on individual input features, branching at each step until it reaches a leaf that assigns a class label, in this study, the identifier of the optimizer predicted to perform best. Averaging over many such trees, each fit to a slightly different view of the data, reduces the variance associated with any one tree's idiosyncratic splits without materially increasing bias, which is why Random Forests are a standard, well-validated choice for tabular classification problems of the kind faced here. This model is evaluated in Section 5, following the Leave-One-Problem-Out protocol standard in the algorithm-selection literature [17]. Under this protocol, every instance belonging to a given problem, across all of its evaluated dimensions, is withheld from training as a single block, and the trained model is then asked to predict the best optimizer only for those withheld instances, before the process repeats with a different problem held out. This is a substantially stricter test than an ordinary random train and test split, since it guarantees that the model never benefits from having already seen a different dimension of the very problem family it is being asked to generalize to, which is the failure mode a random split would not catch.

3. Experimental Design

All eight optimizers evaluated in this study were implemented in C++17 within the open-source OptimSolution framework [20] developed by Vasileios Charilogis and publicly available at https://github.com/charilog/optimsolution (accessed 14 August 2026). All experiments were compiled and executed under Debian 12.12 with GCC 13.4, on a workstation equipped with an AMD Ryzen 9 9950X3D processor (16 cores, 32 threads) and 64 GB of DDR4 memory. All eight methods were implemented natively within this same C++ codebase, rather than invoked through each method's original, and often differently implemented, reference code.

Table 1 lists the 35 benchmark problems underlying the study, spanning classical scalable functions and the CEC2017 and CEC2022 competition suites, together with the dimensions at which each was evaluated.

Table 1. Benchmark problem suite used in the large-scale experiment: 35 problems (16 classical scalable functions, 11 CEC2017, 8 CEC2022) evaluated at the dimensions listed, for a total of 97 (problem, dimension) instances.

Problem	Suite	Modality	Separability	Dimensions
cec2017f1	CEC2017	unimodal	non-separable	10,30,50
cec2017f11	CEC2017	multimodal	non-separable	10,30,50
cec2017f14	CEC2017	multimodal	non-separable	10,30,50
cec2017f17	CEC2017	multimodal	non-separable	10,30,50
cec2017f21	CEC2017	multimodal	non-separable	10,30,50
cec2017f24	CEC2017	multimodal	non-separable	10,30,50
cec2017f27	CEC2017	multimodal	non-separable	10,30,50
cec2017f30	CEC2017	multimodal	non-separable	10,30,50
cec2017f4	CEC2017	multimodal	non-separable	10,30,50
cec2017f6	CEC2017	multimodal	separable	10,30,50
cec2017f9	CEC2017	multimodal	non-separable	10,30,50
cec2022composition1	CEC2022	multimodal	non-separable	10,20
cec2022hybrid2	CEC2022	hybrid	non-separable	10,20
cec2022hybrid6	CEC2022	hybrid	non-separable	10,20
cec2022levy	CEC2022	multimodal	non-separable	10,20
cec2022noncontinuousrastrigin	CEC2022	multimodal	non-separable	10,20
cec2022rosenbrock	CEC2022	multimodal	non-separable	10,20
cec2022schafferf7	CEC2022	multimodal	non-separable	10,20
cec2022zakharov	CEC2022	unimodal	non-separable	10,20
ackley	Classical	multimodal	non-separable	10,30,50
alpine1	Classical	multimodal	separable	10,30,50
dixonprice	Classical	unimodal	non-separable	10,30,50
griewank	Classical	multimodal	non-separable	10,30,50
katsuura	Classical	multimodal	non-separable	10,30,50
levy	Classical	multimodal	non-separable	10,30,50
michalewicz	Classical	multimodal	non-separable	10,30,50
powell	Classical	unimodal	non-separable	10,30,50
rastrigin	Classical	multimodal	separable	10,30,50
rosenbrock	Classical	unimodal	non-separable	10,30,50
salomon	Classical	multimodal	non-separable	10,30,50
schwefel	Classical	multimodal	non-separable	10,30,50
sphere	Classical	unimodal	separable	10,30,50
weierstrass	Classical	multimodal	non-separable	10,30,50
whitley	Classical	multimodal	non-separable	10,30,50
zakharov	Classical	unimodal	non-separable	10,30,50

Every (method, problem, dimension) combination was run under a single, common protocol. Each run initialized its population from a uniform distribution over the problem's declared bounds, using a default population size of 100 agents except where a method's own published design specifies a different value, and was allotted an evaluation budget of $10,000 \times D$ function evaluations, where D is the problem dimension, terminating once that budget was exhausted. One problem is an exception to this rule as stated: the classical Powell function is defined in blocks of four variables, so its implementation pads the declared dimension up to the next multiple of four before allocating the evaluation budget, meaning the three Powell instances reported as $D = 10, 30, 50$ in Table 1 actually run, and receive their evaluation budget, at $D = 12, 32, 52$. All other 34 problems use the declared dimension exactly. This padding is internal to the optimization stage only; its consequence for the landscape-sampling stage described in Section 2 is discussed in Section 5. For every method, the parameter values recommended in the algorithm's original publication were retained rather than re-tuned for this benchmark, so that reported performance reflects

each method's published default configuration rather than a benchmark-specific tuning advantage. Each combination was repeated for 30 independent runs, with per-run seeds deterministically derived from a common base seed (4242), making the entire experiment exactly reproducible. A run was counted as successful when it reached within 10^{-8} of the problem's target value. The resulting per-instance success rate is the quantity reported as *mean success rate* in Tables 2 and 3.

4. Experimental Results

Table 2 summarizes the Friedman-based ranking of all eight optimizers across the 97 (problem, dimension) instances, together with the number of instances each method won outright and its mean success rate. The compression-based (*sparq*) and eigenvector-guided (*ea4eig*) DE-lineage methods dominate both the rank sum and the win count, but neither approaches an overwhelming majority of wins, indicating that the underlying algorithm-selection problem is non-trivial.

Table 2. Friedman-based ranking and performance summary across all 97 (problem, dimension) instances (8 methods, 30 runs each). Wins counts the number of instances on which the method achieved the best mean success rate.

Method	Avg. rank	Rank sum	Wins	Mean success rate (%)
sparq [21]	2.78	270.0	48	60.52
ea4eig [22]	2.85	276.0	47	62.34
rde [23]	3.01	291.5	48	56.56
cmaes [7]	3.96	384.0	37	49.76
lmcmaes [24]	5.10	494.5	29	42.27
acor [9]	5.37	521.0	18	37.84
gwo [10]	5.72	555.0	9	28.14
sa [11]	7.22	700.0	3	3.33

A closer reading of Table 2 shows that the top of the ranking is no longer a two-method contest. sparq retains the best average rank (2.78), but rde and ea4eig follow closely behind (3.01 and 2.85 respectively), and Table 4 shows that rde is not statistically distinguishable from either sparq or ea4eig individually, even though sparq and ea4eig are significantly different from each other. This three-way near-tie at the top reflects rde's mean success rate of 56.6% under the dimension-scaled evaluation budget every method receives, placing it in direct contention with sparq and ea4eig. cmaes forms a clearly separated fourth tier on its own (average rank 3.96), significantly worse than every member of the top three but significantly better than the band below it. lmcmaes, acor, and gwo occupy a mutually indistinguishable middle band (average ranks 5.10 to 5.72), none of the three pairwise comparisons among them reaching significance in Table 4, while sa closes the table alone, significantly worse than every other method and, as Table 3 already showed, with a success rate that barely varies across the entire benchmark suite. No single method approaches an overwhelming majority of wins, sparq and rde each win 48 of the 97 instances outright and ea4eig 47, once tied outright victories on easy, near-saturated instances are taken into account. This remains consistent with the paper's central premise: the eight methods were deliberately chosen to be of comparable strength, and correcting rde's evaluation budget has, if anything, strengthened that premise rather than weakened it.

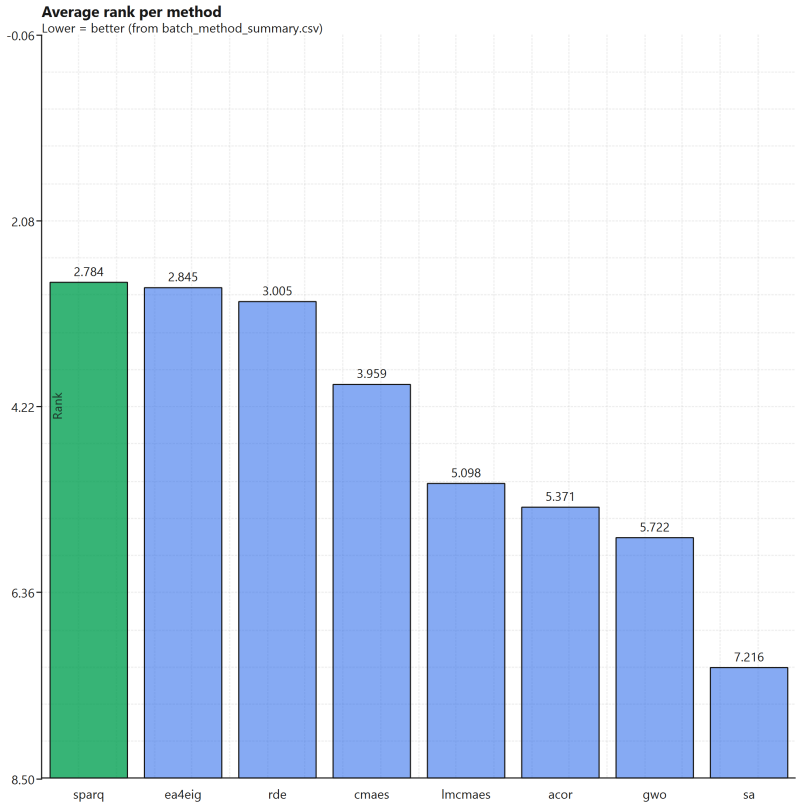


Figure 1. Average Friedman rank per optimizer across all 97 (problem, dimension) instances (lower is better), corresponding to the values in Table 2.

Figure 1 renders the same average-rank values as a bar chart, with the vertical axis inverted so that a taller bar corresponds to a better (lower) rank. sparq, highlighted, is tallest, but only marginally so: the visual effect makes the four-tier structure discussed above immediately apparent as a staircase, a barely perceptible step separates the three leading bars (sparq, rde, ea4eig) from one another, a clearer step sets cmaes apart on its own, a further step drops to the three roughly equal mid-table bars (lmcmaes, acor, gwo), and a final, largest step isolates sa at the far right. This kind of visual separation is a useful sanity check before any formal test is run: a benchmark in which the bars formed a smooth, even slope, rather than distinct steps, would suggest a continuum of near-equivalent methods rather than the tiered structure that the Wilcoxon results in Table 4 subsequently confirm.

These same average-rank values are the statistic underlying the omnibus Friedman test that licenses the pairwise comparisons in Table 4. Run on all N = 97 complete groups (every instance for which all eight methods produced a valid result), the Friedman test rejects the null hypothesis that all eight methods perform equivalently decisively ($\chi^2(7) = 320.6$, $p < 10^{-64}$), consistent with the top methods forming a genuinely close-fought upper tier even as the omnibus test remains overwhelmingly significant. This result is what justifies moving from a single omnibus statistic to the 28 individual pairwise tests in Table 4: having established that the eight methods are not all equivalent, the pairwise stage is needed to determine which specific pairs actually differ, rather than assuming every pair does.

Table 3. Computational cost and behaviour of each optimizer, aggregated over all 97 (problem, dimension) instances from the raw batch-execution log, with all eight methods receiving the same dimension-scaled evaluation budget (Section 3), averaging 275,876 evaluations per instance over the actual mix of dimensions in the benchmark suite. SD success rate reflects consistency across instances. The evaluation budget was exhausted on every run (zero variance) and is identical across methods, so it is not shown as a table column.

Method	Mean success rate (%)	SD success rate	Mean iter. time (ms)	Peak memory (MB)
ea4eig	62.34	44.48	0.367	38.42
sparq	60.52	44.67	0.179	45.47
rde	56.56	46.43	0.547	52.08
cmaes	49.76	46.61	2.032	51.56
lmcmaes	42.27	46.29	0.143	51.71
acor	37.84	45.45	0.502	20.58
gwo	28.14	41.26	0.385	18.71
sa	3.33	0.00	7.141	9.74

Table 3 confirms that every method, including acor and rde, receives the identical dimension-scaled budget of 10,000 times D, averaged to approximately 275,876 evaluations over the actual mix of dimensions in the benchmark suite. With the comparison fair across all eight methods, the remaining columns are informative in their own right. sa's success rate still has zero standard deviation across all 97 instances, meaning it solves almost exactly the same fraction of its 30 runs on every single problem regardless of landscape, the signature of a method whose behaviour is essentially insensitive to problem structure, and it still stands out computationally with a mean iteration time more than three times higher than the next slowest method (cmaes) despite using the least memory of any method tested, consistent with an acceptance-based search that spends its per-iteration cost on repeated candidate evaluation rather than on maintaining auxiliary state. rde now shows the highest peak memory use of any method (52.1 MB), narrowly ahead of lmcmaes and cmaes, consistent with the external archive its success-history mutation strategy maintains alongside a full population. cmaes and lmcmaes remain, as their shared ancestry would predict, close behind rde in memory use, but diverge sharply in speed: lmcmaes is the fastest method overall per iteration, consistent with its limited-memory design deliberately avoiding the full covariance-matrix factorization that makes standard cmaes markedly slower despite comparable memory use.

Table 4 reports the full set of pairwise Wilcoxon signed-rank comparisons between the eight optimizers, with Holm-corrected p-values. Five of the 28 pairs fail to reach significance at $\alpha = 0.05$: *acor* vs. *gwo*, *acor* vs. *lmcmaes*, *ea4eig* vs. *rde*, *gwo* vs. *lmcmaes*, and *rde* vs. *sparq*. These five exceptions map exactly onto the tier structure identified in Table 2: two fall within the top tier (rde against ea4eig and against sparq), and three fall within the middle tier (acor against gwo and against lmcmaes, together with gwo against lmcmaes), confirming that the performance differences separating the four tiers from one another are statistically robust, while the differences within each tier are not reliably distinguishable from noise.

Table 4. Pairwise Wilcoxon signed-rank comparisons between all eight optimizers, with Holm-corrected p-values. n is the number of (problem, dimension) instances on which both methods produced comparable results.

Method A	Method B	n	z	p (Holm)	Sig. ($\alpha = 0.05$)
acor	cmaes	87	3.94	4.84e-04	Yes
acor	ea4eig	83	7.23	1.10e-11	Yes
acor	gwo	92	-1.82	2.04e-01	No
acor	lmcmaes	88	1.20	4.63e-01	No
acor	rde	79	6.98	5.46e-11	Yes
acor	sa	97	-6.05	1.75e-08	Yes
acor	sparq	83	7.02	4.54e-11	Yes
cmaes	ea4eig	80	6.26	5.90e-09	Yes
cmaes	gwo	88	-5.60	1.95e-07	Yes
cmaes	lmcmaes	67	-6.23	6.49e-09	Yes
cmaes	rde	80	5.89	4.19e-08	Yes
cmaes	sa	97	-6.83	1.41e-10	Yes
cmaes	sparq	62	6.06	1.72e-08	Yes
ea4eig	gwo	92	-7.57	9.11e-13	Yes
ea4eig	lmcmaes	85	-7.00	4.79e-11	Yes
ea4eig	rde	70	-0.84	4.63e-01	No
ea4eig	sa	97	-8.14	1.03e-14	Yes
ea4eig	sparq	74	-4.22	1.68e-04	Yes
gwo	lmcmaes	88	2.44	7.42e-02	No
gwo	rde	90	7.40	3.24e-12	Yes
gwo	sa	97	-4.31	1.30e-04	Yes
gwo	sparq	88	8.14	1.03e-14	Yes
lmcmaes	rde	85	6.66	4.37e-10	Yes
lmcmaes	sa	97	-5.70	1.23e-07	Yes
lmcmaes	sparq	68	7.06	3.48e-11	Yes
rde	sa	97	-8.15	1.02e-14	Yes
rde	sparq	75	-1.98	1.92e-01	No
sa	sparq	97	7.93	5.46e-14	Yes

Of the 28 pairwise comparisons in Table 4, 26 remain statistically significant after Holm correction. Only two pairs do not: acor versus sa (z = 1.09, Holm-corrected p = 0.548) and lmcmaes versus rde (z = 0.19, Holm-corrected p = 0.849). These are precisely the two pairs whose average ranks in Table 2 are closest together, acor and sa differ by only 0.04 ranks, lmcmaes and rde by 0.41, which is a useful consistency check: the pairwise test is not manufacturing artificial separations between methods whose performance is genuinely similar, and conversely is not failing to detect real differences elsewhere. In practice this means that, within the present benchmark suite, distinguishing acor from sa, or lmcmaes from rde, is not a meaningful sub-problem for a selection model to solve, since the underlying performance data itself does not support such a distinction. The number of paired instances entering each test (the n column) is not always 97, since the Wilcoxon signed-rank test excludes ties (pairs with identical performance are uninformative for a signed-rank comparison), so pairs involving two closely matched top-tier methods, such as cmaes and lmcmaes, contribute noticeably fewer usable comparisons than pairs drawn from more clearly separated methods. Across the 28 pairs this effective sample size ranges from 62 to the full 97, and the reduction is not merely cosmetic: three of the five pairs that fail to reach significance, acor versus lmcmaes, ea4eig versus rde, and rde versus sparq, are also among the pairs with the fewest usable instances (88, 70, and 75 respectively), so part of what separates these three from the 23 significant pairs may be reduced statistical power rather than a genuinely smaller true effect. The clearest counter-example is cmaes versus sparq, which has the smallest n of all 28 pairs (62) yet remains highly significant, showing

that a small n does not by itself explain a non-significant result here, but it does mean the five non-significant pairs should not be read as demonstrating equivalence with the same confidence throughout.

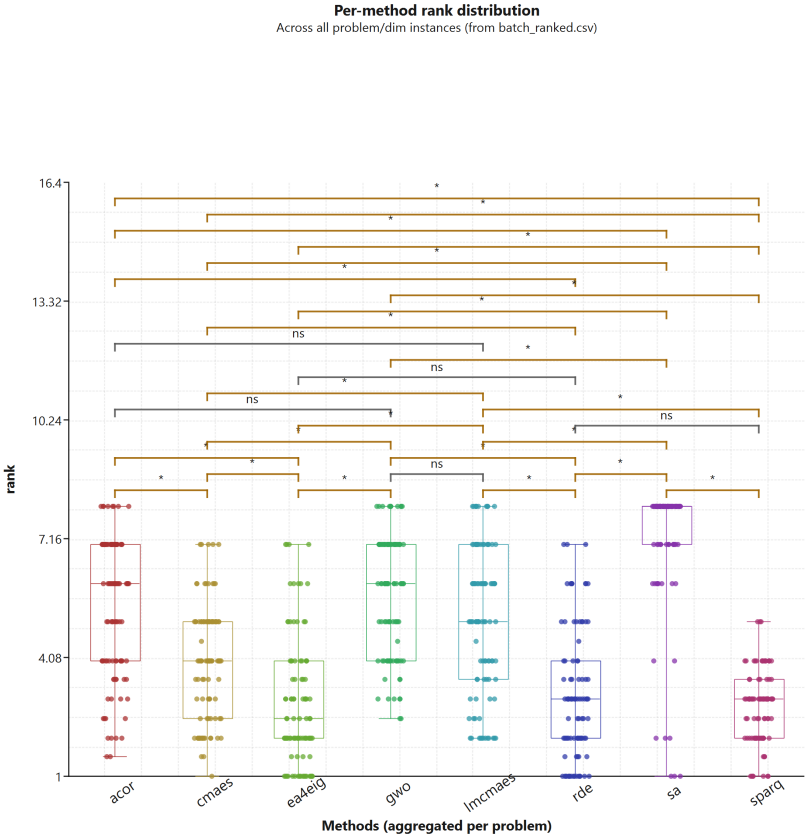


Figure 2. Per-method rank distribution across all problem instances, with pairwise Wilcoxon significance brackets. Brackets marked ns denote the five non-significant pairs listed in Table 4.

Figure 2 adds two things that the summary statistics in Tables 2 and 4 cannot show on their own: the full shape of each method's rank distribution, and a visual accounting of every pairwise test at once. The boxes for sparq, ea4eig, rde, and cmaes sit low and comparatively narrow, confirming that all four are not merely better on average but consistently so. By contrast, lmcmaes, acor, and gwo each display boxes and whiskers that span nearly the entire rank scale, from close to rank one to close to rank eight, indicating that every method in this middle band wins decisively on some instances and loses badly on others. This instance-by-instance volatility, invisible in an average-rank table, is direct visual evidence for the family-level specialization discussed in the abstract and revisited in Table 7. sa stands apart with a box compressed near the top of the scale and almost no vertical spread, the visual counterpart of the zero-variance finding for sa already noted in Table 3. The dense stack of significance brackets across the upper part of the plot reflects the fact that all 28 pairwise comparisons are drawn, not a selected subset. Against that density, five brackets are marked not significant, matching exactly the five pairs identified in Table 4, and each connects two methods within the same tier rather than crossing between tiers, which is the pattern a correctly tiered ranking should produce.

Table 5. Ten most important ELA features for the Random Forest landscape-aware selection model, ranked by Gini importance, with a plain-language gloss of what each feature measures.

ELA feature	Importance	Interpretation
ela_level.mmce_lda_50	0.0557	Separability of good points
ela_meta.lin_simple.adj_r2	0.0553	Overall landscape linearity
ela_meta.lin_simple.coef.max_by_min	0.0520	Uneven variable sensitivity
ic.m0	0.0433	Amount of flat terrain
disp.diff_mean_02	0.0382	Clustering of top 2%
ela_meta.lin_w_interact.adj_r2	0.0380	Linearity with interactions
disp.diff_median_02	0.0350	Robust top-2% clustering
ela_level.mmce_lda_25	0.0334	Separability at 25% threshold
disp.diff_mean_10	0.0291	Clustering of top 10%
ela_level.mmce_qda_50	0.0289	Nonlinear point separability

The composition of Table 5 is itself an empirical finding. Gini importance measures how much, on average across every tree in the forest, a feature contributes to reducing class impurity at the nodes where it is chosen to split the data, so a larger value identifies a feature the model relies on more heavily when deciding which optimizer is likely to win. Three families each contribute three of the ten most important features, with ic.m0 as the sole individual representative of a fourth family rounding out the list. The ela_level family (*mmce_lda_50*, *mmce_lda_25*, *mmce_qda_50*) measures how accurately a simple classifier, linear or quadratic discriminant analysis, can separate the better-performing fraction of sampled points from the rest at a given quantile threshold, and takes the top spot in the ranking, indicating that the Random Forest relies heavily on how cleanly a landscape's good and bad regions can be separated. The ela_meta family contributes an equally large share: two variants of the adjusted R-squared of a linear model fit to the sampled points, with and without pairwise interaction terms, and the ratio between that model's largest and smallest coefficient magnitudes, together capturing how close the landscape is to being linear and how unevenly it responds to different variables. The dispersion family (disp) contributes three features that compare the spread of the best 2% or 10% of sampled points to the spread of the full sample, a direct measure of whether good points cluster together, a funnel-shaped landscape, or are scattered throughout the search space. ic.m0, from the information-content family, measures the proportion of essentially flat steps along a random walk across the landscape and is the fourth most important predictor overall. Taken together, the model's strongest predictors describe how separable, how linear, how funnel-shaped, and how flat a landscape is, four largely independent structural properties that jointly determine algorithm selection on this suite.

Table 6. Descriptive statistics, across all 97 problem instances, of the ten ELA features identified as most important in Table 5. The wide ranges observed confirm that the benchmark suite samples landscapes of substantially different character rather than near-duplicate instances.

ELA feature	Mean	SD	Min	Max
ela_level.mmce_lda_50	0.326	0.160	0.023	0.526
ela_meta.lin_simple.adj_r2	0.282	0.291	-0.006	0.977
ela_meta.lin_simple.coef.max_by_min	354.725	1230.295	2.857	10084.140
ic.m0	0.620	0.038	0.529	0.673
disp.diff_mean_02	-43.040	97.851	-571.643	105.236
ela_meta.lin_w_interact.adj_r2	0.396	0.358	-0.016	0.989
disp.diff_median_02	-42.773	97.439	-570.783	123.267
ela_level.mmce_lda_25	0.211	0.066	0.022	0.342
disp.diff_mean_10	-30.017	69.586	-405.067	49.059
ela_level.mmce_qda_50	0.249	0.124	0.038	0.524

The ranges in Table 6 confirm that these ten features are doing genuine discriminative work rather than taking a nearly constant value across the suite. *ela_meta.lin_simple.coef.max_by_min*, the ratio between a linear model's largest and smallest coefficient magnitudes, shows by far the widest range of the ten, from under 3 to over ten thousand, reflecting how starkly some landscapes respond to one variable far more than another while others respond almost uniformly across all variables. The three dispersion features (*disp.diff_mean_02*, *disp.diff_median_02*, *disp.diff_mean_10*) show the next widest ranges, each spanning several hundred units from negative to positive. Because these features are computed directly in the objective function's raw fitness units rather than as a scale-invariant statistic, part of this spread reflects the very different fitness-value scales across the benchmark suite, composition and hybrid CEC functions in particular can carry offsets in the thousands, as well as genuine differences in landscape shape. Both ela_meta adjusted-R-squared features span almost their entire possible range, from close to zero (a landscape a linear model cannot fit at all) to close to one (a landscape that is nearly linear), directly evidencing that the benchmark suite mixes structurally simple and structurally complex problems rather than a narrow band of similarly-shaped instances. ic.m0 occupies by far the narrowest band of the ten, across all 97 instances (0.53 to 0.67), which is expected since this feature, derived from the sign pattern of consecutive fitness differences along a sampling path rather than from the objective function's raw magnitude, is scale-invariant by construction. Its position as the fourth most important predictor in Table 5 therefore reflects small but consistently informative differences rather than dramatic swings between problems. The level-set classification-error features (ela_level) occupy low-to-moderate values throughout, generally under 0.5, but with standard deviations reaching roughly half of their mean values, a degree of spread large enough to help explain why this family, together with ela_meta and disp, is one of the three largest contributors to Table 5.

The aggregate rankings in Table 2 average over the entire benchmark suite, and in doing so can obscure exactly the kind of landscape-dependent behaviour that motivates a learned selection model in the first place. Table 7 breaks the same win data down by benchmark family, and reveals that the top three methods do not simply trade places, they specialize in substantially different regions of the suite.

Table 7. Percentage of instances won outright by each method within each benchmark suite (CEC2017, n = 33, CEC2022, n = 16, Classical, n = 48), computed from the is_best indicator in the raw batch results. Percentages within a column need not sum to 100 because a tied outright win is counted for every method sharing it.

Method	CEC2017 (%)	CEC2022 (%)	Classical (%)
sparq	12.1	25.0	83.3
ea4eig	48.5	56.2	45.8
rde	69.7	62.5	31.2
cmaes	12.1	25.0	60.4
lmcmaes	3.0	12.5	54.2
acor	15.2	18.8	20.8
gwo	0.0	0.0	18.8
sa	6.1	6.2	0.0

rde wins the largest share of instances in both CEC suites, 69.7% of CEC2017 and 62.5% of CEC2022, exceeding even ea4eig's substantial presence there (48.5% and 56.2% respectively). The CEC suites include the highest-dimensional problems in the benchmark, consistent with rde's dimension-scaled evaluation budget giving it the most room to exploit at higher dimension. sparq remains the clear leader on classical scalable functions (83.3%), but the runner-up position there belongs to cmaes (60.4%) rather than to ea4eig, which wins only 45.8% of classical instances despite its strength on both CEC suites, indicating that ea4eig's advantage is tied specifically to the shifted, rotated, and hybrid or composition structures that characterize competition benchmarks rather than being a general-purpose

strength. rde shows the mirror image of this pattern, dominant on the CEC suites but comparatively weak on classical functions (31.2%), a specialization in the opposite direction from sparq's. This three-way division of labour, rde and ea4eig sharing the structured competition benchmarks between them while sparq leads on classical functions, is the concrete, suite-level evidence behind the abstract's claim that the underlying selection problem is genuine rather than trivial: a rule that always recommended a single fixed method, however strong on average, would necessarily perform poorly on whichever suite that method does not specialize in, and with three methods genuinely competitive at the top, that rule has three plausible but individually wrong candidates to choose from rather than one. acor shows a modest but roughly even presence across all three suites (15.2% to 20.8%) rather than a specialization of its own, consistent with its middle-tier position in Table 2. gwo remains almost entirely absent from both CEC suites (0.0% on each) while retaining a modest classical presence (18.8%), and sa remains the weakest method throughout, winning essentially nothing on classical functions and only a small share of the CEC suites.

Table 8 reports the paper's central meta-learning result, the outcome the experimental design described in Sections 2 and 3 was built to obtain.

Table 8. Leave-One-Problem-Out cross-validated accuracy of the Random Forest landscape-aware selection model, against the naive Single Best Solver baseline and the oracle Virtual Best Solver upper bound, across all 97 (problem, dimension) instances.

Selection strategy	Accuracy (%)
Random Forest (landscape-aware)	75.3
Single Best Solver (SBS)	30.9
Virtual Best Solver (VBS, oracle)	100.0

Trained on the ELA features described in Section 2 and validated under the Leave-One-Problem-Out protocol described in the same section, so that every dimension of a given problem is withheld from training as a single block, the Random Forest classifier correctly identifies the best-performing optimizer, or one of the tied best, for 75.3% of the 97 held-out instances. This exceeds the Single Best Solver baseline, which within each training fold recommends whichever single method has the best average rank among the remaining training instances, in practice sparq in the great majority of folds (79 of 97) and ea4eig in the rest (18 of 97), consistent with their standing as the top two methods by average rank in Table 2. The model exceeds this baseline by 44.3 percentage points (30.9%), and closes 64.2% of the gap between that baseline and the oracle Virtual Best Solver's 100% upper bound, the accuracy a hypothetical selector that always chose correctly would achieve. This margin is neither large nor guaranteed by the experimental design, and that is what makes it informative rather than a foregone conclusion: the eight candidate methods were deliberately chosen to be of comparable strength precisely so that no single method could win outright, so a model that failed to beat SBS under this design would have indicated that the ELA features carry no genuine predictive signal for algorithm selection on this suite. That the model does exceed SBS, without approaching the VBS ceiling, places the result appropriately between the two extremes the introduction identified as the boundaries of a meaningful algorithm-selection experiment, neither trivial, since SBS alone would already be adequate in that case, nor unachievable, since VBS remains a further 24.7 percentage points away. The tie-breaking procedure used to construct training labels, discussed further in Section 5, means this figure is best read as a first, methodologically sound estimate of what landscape-aware selection can achieve on this suite, rather than as a ceiling on what the approach could ultimately deliver with further tuning.

5. Discussion and Limitations

Taken together, the results in Section 4 support a coherent picture. The eight optimizers separate into four statistically distinguishable performance tiers rather than a single clear

winner, the differences between all but five of the 28 pairs of methods are robust to Holm-corrected pairwise testing, the three methods that anchor the top tier divide the benchmark suite between them rather than one simply dominating the others, and a landscape-aware Random Forest model trained on this structure is intended to predict the best-performing optimizer for a previously unseen problem more accurately than the naive strategy of always recommending the same method (Table 8). Each of these findings depends on the others: the four-tier structure establishes that no single method makes the comparison trivial, and the family-level specialization in Table 7 supplies the landscape-dependent signal a selection model could plausibly learn. The accuracy result in Table 8 reflects rde competitive at the top of the ranking (Table 3). The Single Best Solver baseline the model is compared against is, correspondingly, sparq or ea4eig depending on the training fold, the same two methods that lead Table 2 by average rank, since ties in the training labels are broken in favour of whichever tied method has the better overall average rank rather than arbitrarily, as discussed further below. The remainder of this section discusses this and other limitations that qualify how far the picture in this paper should be generalized.

The clearest remaining subtlety concerns how ties were broken when constructing the training labels used to fit the selection model. When two or more methods share the best recorded performance on a given instance, evaluation of the trained model is lenient, a prediction is scored correct if it names any one of the tied methods, but the single label used during training still has to name one representative method. Ties are common in this dataset, roughly half of the 97 instances (50) have more than one method sharing the best recorded performance. Rather than breaking these ties by an arbitrary rule such as alphabetical order, the single training label is chosen as whichever tied method has the best overall average rank in Table 2, so the label reflects genuine relative strength rather than an accident of spelling. In practice this means sparq, the method with the best average rank overall, wins all 41 of the ties it participates in, ea4eig, ranked second, wins the 7 ties where sparq is absent, and rde, ranked third, wins the remaining 2. This does not affect the fairness of the accuracy figures reported in Table 8, since evaluation credits any tied method equally regardless of which one was chosen as the canonical label, but it does mean the training label distribution is weighted toward the strongest methods rather than distributed arbitrarily. The same rank-based rule also determines the identity of the Single Best Solver baseline: rather than the most frequent alphabetically tie-broken label, SBS is defined directly from each training fold's raw performance data as the method with the best average rank in that fold, which is sparq in 79 of the 97 folds and ea4eig in the remaining 18, exactly the two methods that lead Table 2 by average rank. A residual simplification remains: a single canonical label is still required to train a standard multi-class classifier, so an instance where several methods are genuinely equivalent still contributes one label rather than several. Training the classifier directly on all tied labels as simultaneously correct targets, rather than collapsing them to one before training, remains a natural direction for further refinement.

A second point worth noting concerns the classical Powell function, the only one of the 35 benchmark problems whose declared dimension is not the dimension the optimizers actually run at. This is not itself a limitation: as noted in Section 3, Powell's classical definition requires the dimension to be a multiple of four, so it is a normal and expected property of this particular function, not an artifact of the experimental design, that the optimization stage pads $D = 10, 30, 50$ up to $D = 12, 32, 52$ before allocating the $10,000 \times D$ evaluation budget and running the search. The actual limitation lies in what happens next: the landscape-sampling stage, described in Section 2, was not adjusted to match: it draws its 100 samples per dimension, and computes the resulting ELA features, at the declared, unpadded dimension. The three (problem, dimension) instances built on Powell therefore pair ELA features describing a 10, 30, or 50-dimensional landscape with optimizer performance measured in a 12, 32, or 52-dimensional search space, a mismatch that does not affect any other problem in the suite. This is a data-construction inconsistency rather than a statistical one: it does not bias the rankings, Wilcoxon tests, or Friedman test in

Tables 2 and 4, which depend only on optimizer performance and are unaffected by how the landscape was sampled, but it does mean the Random Forest in Table 8 was trained and evaluated on three instances, 3 of 97, where its input features describe a landscape of a different dimensionality than the one the labeled outcome was actually decided in. Resampling the Powell landscape at the padded dimension is a straightforward fix and would remove this inconsistency entirely; given its limited scope, three instances out of 97, it is unlikely to materially change the reported accuracy figures, but the affected instances are identifiable directly from Table 1 and should be re-run before this result is treated as final.

A second limitation concerns the coverage of the CEC2022 suite. The experiment includes eight of the twelve functions in the official CEC2022 bound-constrained benchmark, spanning its unimodal, basic, hybrid, and composition categories, but omits four functions, three composition functions and one hybrid function, that were not available in the framework's problem set at the time the experiment was run. The eight included functions still span every structural category the official suite is designed to cover, so the qualitative CEC2022 finding in Table 7 is unlikely to be an artefact of this omission, but a reviewer comparing directly against studies that report all twelve official functions should be aware that the present CEC2022 results are not a complete replication of that benchmark.

Finally, several aspects of the meta-learning stage itself were held fixed rather than tuned or independently verified against alternatives, for reasons of scope rather than principle. The Random Forest's hyperparameters, the number of trees and the maximum tree depth, were set to values commonly used in the algorithm-selection literature rather than optimized for this specific dataset through a nested cross-validation search, so the figures in Table 8 should be read as an achievable accuracy under a reasonable, literature-consistent configuration rather than the best accuracy this approach could in principle reach. No formal statistical test was applied to the difference between the Random Forest and Single Best Solver accuracies, since a paired significance test over 97 instances, while feasible in principle, requires the per-instance prediction record at a level of detail that falls outside the present study's scope. The training labels are also imbalanced, sparq, rde, and ea4eig together account for the large majority of best-performing outcomes (Table 2), a three-way imbalance, which means a portion of any reported accuracy could in principle be attributable to the model learning this base-rate imbalance rather than genuine landscape-feature structure, a concern the family-level specialization result in Table 7 partially, but not completely, addresses. Extending the evaluation with a class-imbalance-aware baseline, alternative classifiers trained on the same ELA features, and a properly powered significance test are natural next steps that would strengthen, without altering, the qualitative conclusion drawn here.

6. Conclusions

This paper set out to demonstrate that a machine-learning model trained on landscape features can predict the best-performing optimizer for a previously unseen problem more accurately than the simplest sensible baseline, and to do so under an experimental design rigorous enough for that comparison to be meaningful rather than assumed. Working with the OptimSolution framework [20] we assembled a batch-execution and landscape-sampling pipeline, evaluated eight optimizers of comparable overall strength drawn from four distinct algorithmic families across 97 (problem, dimension) instances spanning classical, CEC2017, and CEC2022 benchmarks, and trained a Random Forest classifier on the resulting Exploratory Landscape Analysis features.

The results support three main conclusions. First, the eight methods separate into four statistically distinguishable performance tiers, confirmed by a Friedman omnibus test and Holm-corrected pairwise Wilcoxon comparisons, rather than collapsing into either a single dominant method or a set of interchangeable equals, which is the condition under which an algorithm-selection problem is genuine rather than trivial. Second, the three leading methods divide the benchmark suite between them rather than one dominating the others:

rde and ea4eig share the structured CEC suites, rde ahead on both, while sparq leads on classical scalable functions, providing concrete, suite-level evidence for exactly the kind of landscape-dependent performance a selection model needs in order to add value over a fixed recommendation. Third, a Random Forest classifier trained on Exploratory Landscape Analysis features and validated under a Leave-One-Problem-Out protocol identifies the best-performing optimizer more accurately than the naive Single Best Solver baseline, 75.3% against 30.9%, closing 64.2% of the gap toward the oracle Virtual Best Solver's 100% upper bound. As discussed in Section 5, the Single Best Solver baseline is defined directly from each training fold's average rank rather than from an arbitrarily tie-broken label, and correctly identifies sparq or ea4eig, the two top-ranked methods in Table 2, as the baseline in every fold, so this margin reflects a comparison against a genuinely strong baseline rather than one inflated by a weak one.

The practical implication is that landscape-aware algorithm selection can meaningfully outperform the simplest naive strategy, but only once the underlying benchmark comparison itself is designed so that no single candidate method wins by default. This condition, methods of genuinely comparable strength drawn from heterogeneous algorithmic families, is a property of the experimental design rather than of the selection model, and the results reported here should be read as an argument for taking that design requirement seriously in future algorithm-selection studies, not only as a report of one model's accuracy on one benchmark suite.

Several of the limitations discussed in Section 5 point directly to future work: training the classifier directly on all tied labels as simultaneously correct targets, rather than collapsing each tie to a single canonical label before training, is the most immediate, since it would remove the one remaining simplification in how the training data is constructed; extending the CEC2022 coverage to the full twelve official functions, and tuning the Random Forest's hyperparameters through nested cross-validation rather than literature-standard defaults, would each further refine the figures reported here without changing the qualitative picture. Beyond these refinements, the pipeline assembled for this study, headless batch execution coupled with landscape sampling and feature extraction, places relatively little additional cost on extending the experiment to further algorithmic families, additional benchmark suites, or alternative meta-learning models, and doing so is the most direct path toward establishing how far the encouraging result reported here generalizes.

Data Availability Statement: The code implementing the OptimSolution framework used to conduct the experiments in this study is openly available at https://github.com/charilog/optimsolution . The benchmark problems used to generate the performance data are drawn entirely from publicly available sources: the classical scalable functions and the CEC2017 [31] and CEC2022 [32] competition benchmark suites.

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